Suppression of the quadrupole-misalignment false EDM signal by orbit-based methods in a proton EDM storage ring

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In the storage ring proposed for measuring the proton electric dipole moment (EDM), quadrupole misalignments of a few microns generate, through a geometric (Berry) phase, a vertical spin-precession signal that mimics a genuine EDM and exceeds the target sensitivity of 10^{-29} $e$cm by up to three orders of magnitude. Using symplectic spin tracking, we examine how far this false signal can be removed by measuring and correcting the beam orbit. We find that standard orbit correction alone does not guarantee suppression: the misalignment pattern separates into an orbit-visible antisymmetric component and a symmetric component to which the orbit response is nearly blind (up to two orders of magnitude weaker mode by mode), and the false EDM is a bilinear functional of the two transverse misalignment patterns, so that it cannot be characterized by a single rms alignment tolerance. Null-finding beam-based alignment reaches the symmetric component that response-matrix inversion cannot; combined with orbit correction, it suppresses the false EDM from several hundred times the target to the target level in our simulations. We quantify the remaining requirements: the fidelity of the steering optics model, the re-alignment cadence against alignment drift, and the stability of the magnetic centers under gradient modulation, which is not yet modeled.

I. INTRODUCTION

A. The frozen-spin method and the false-EDM problem

A permanent electric dipole moment (EDM) of the proton would violate time-reversal symmetry, and its observation above the Standard-Model expectation ($\lesssim 10^{-31}$ $e$cm) would constitute direct evidence of new sources of CP violation. The proposed storage-ring experiment [1] targets a sensitivity of $d = 10^{-29}$ $e$cm using the *frozen-spin* method: protons are stored at the "magic" momentum ($p \approx 0.7$ GeV/c), at which the spin remains aligned with the momentum as the beam circulates. An EDM then reveals itself as a slow rotation of the spin out of the ring plane, driven by the radial electric bending field. The measured quantity is the out-of-plane precession rate,

$$f \equiv \frac{dS_y}{dt}, \qquad d = 10^{-29}\, e\!\cdot\!\mathrm{cm} \;\leftrightarrow\; f \approx 1 \text{ nrad/s}. \tag{1}$$

Any competing effect that rotates the spin out of plane at this level must be eliminated or bounded; such effects are referred to as *false EDM* signals.

The subject of this paper is the false EDM generated by transverse misalignments of the focusing quadrupoles. A quadrupole has zero field on its magnetic axis, so a beam passing off-axis experiences a field proportional to the offset. A vertical offset dy exposes the beam to a radial magnetic field, which precesses the spin about the radial axis and can therefore tilt it out of the ring plane. A horizontal offset dx exposes the beam to a vertical field, which precesses the spin about the vertical axis; this precession is confined to the ring plane and, like the in-plane precession of the frozen-spin condition itself, does not accumulate in S_y. Either precession alone is nearly harmless: averaged around the ring, its out-of-plane effect largely cancels. When both are present, however, the two precessions do not commute — precessing first about the radial axis and then about the vertical axis is not the same operation as the reverse — and the residue of this non-commutativity accumulates turn after turn as a net rotation about the third axis, which is precisely the out-of-plane direction in which the EDM is sought. This is a geometric (Berry) phase, and its leading dependence is

$$f_{\text{false}} \;\propto\; dx \cdot dy \;\;\Longrightarrow\;\; f_{\text{false}} \propto \sigma^2 \tag{2}$$

for misalignments of rms size σ. We verify the quadratic scaling directly in Sec. II B (measured exponent $p = 2.00 \pm 0.01$, in agreement with Fig. 9(a) of Ref. [1]). At a realistic alignment of $\sigma = 10$ μm the false EDM is of order a thousand times the target of Eq. (1), and it appears in the same observable, dS_y/dt, as the genuine signal.

B. The ring

The lattice studied here is the symmetric-hybrid ring of Ref. [1]: electric bending with magnetic focusing. The ring ($R_0 = 95.49$ m) consists of 24 identical FODO cells. Each cell contains a horizontally focusing quadrupole (QF) and a horizontally defocusing quadrupole (QD) — both *magnetic*, with gradients of equal magnitude ($g \approx 0.2$ T/m) and opposite sign — separated by field-free drifts and by the *electric* bending deflectors:

$$\begin{gathered}\text{QF} \to \text{drift} \to \text{deflector} \to \text{drift} \to \text{QD}\\ \to \text{drift} \to \text{deflector} \to \text{drift} \to (\text{QF}\ldots)\end{gathered}$$

The deflectors are cylindrical electrostatic capacitors that bend the beam in the horizontal plane; at the magic momentum their radial field also freezes the in-plane spin precession. Two properties of this arrangement matter for what follows. First, between any two neighboring quadrupoles the beam traverses a deflector: the deflectors bend and weakly focus the beam *horizontally*, while in the *vertical* plane they act, to a very good approximation, as field-free drifts. Second, the alternating QF/QD gradients make the two transverse planes behave identically in the idealized lattice, with betatron tunes $Q_x \approx Q_y \approx 2.3$ (the tune is the number of transverse oscillations per revolution).

C. Orbit-based alignment as the natural remedy

Since the false EDM originates in the beam–quadrupole offsets, the natural strategy is to measure and correct them using the beam itself. This is standard accelerator practice. The closed orbit — the periodic trajectory about which the stored beam oscillates — is read continuously by beam position monitors (BPMs), located in our model at each of the 48 quadrupoles. Misaligned quadrupoles deflect the closed orbit, and the deflections are related to the offsets through the orbit response matrix; correcting the orbit with steering elements, guided by the singular-value decomposition of that matrix, has been routine since the work of Chung, Decker, and Evans [3]. Beyond orbit correction, the accelerator community locates individual quadrupole centers by *beam-based alignment* (BBA): the quadrupole gradient is varied and the beam position at which the orbit stops responding marks the magnetic center, a procedure that has been accelerated with ac excitation [5] and parallelized over many magnets [6]. Related tools include orbit-response modeling in the presence of degeneracy [4], correction schemes exploiting lattice symmetry [7], and on-line response-matrix refinement [8]. None of the measurement techniques examined in this paper is new in itself; building on the ring design and systematic-error analysis of Ref. [1], we quantify how these standard tools perform on the specific misalignment component that drives the false EDM in this ring, and what alignment program they add up to — an answer that turns out to be structured.

D. Findings and outline

Three results organize the paper.

First, *standard orbit correction alone does not guarantee suppression of the false EDM* (Sec. III). The misalignment pattern separates into two components with sharply different orbit visibility: an *antisymmetric* component (neighboring QF and QD displaced oppositely), which produces a large orbit distortion and is removed by orbit correction, and a *symmetric* component (QF and QD displaced together), whose orbit signature is strongly suppressed — by up to two orders of magnitude mode by mode — and which therefore survives correction. The false EDM is a bilinear functional of the horizontal and vertical misalignment patterns, and the surviving symmetric–symmetric term sets the post-correction floor. An immediate corollary is that the false EDM cannot be characterized by a single rms alignment tolerance: at fixed σ its value varies by more than an order of magnitude with the symmetric/antisymmetric content of the pattern.

Second, *the symmetric component cannot be recovered from orbit readings by response-matrix inversion or amplitude-readout techniques* (Sec. IV), for reasons that are coherent and therefore independent of BPM technology. The counter-rotating beam separation of Ref. [1] shares this property of the orbit.

Third, *null-finding BBA, followed by a final orbit correction, does reach the symmetric component* (Sec. V). BBA locates each magnetic center by a zero-crossing rather than by inverting a matrix, and is therefore not subject to the visibility gap; the final orbit correction removes the antisymmetric residue that BBA leaves behind. In our tracking simulations the combination suppresses the false EDM from several hundred times the target to the target level, where each operation alone stalls an order of magnitude or more above it.

These results fix the place of orbit correction within the alignment program of Ref. [1], which combines BBA with a four-fold measurement — the counter-rotating difference repeated at reversed quadrupole polarity — that cancels the magnetic-lattice false EDM exactly when the machine is identical between the two polarity states. We find, first, that orbit correction cannot substitute for BBA, because the symmetric component is invisible to the orbit in every configuration; and second, that added to BBA it can substitute for the polarity reversal, removing the antisymmetric channel continuously and without a second pass of the experiment, while BBA supplies the symmetric alignment that the polarity cancellation itself requires to survive drift (Sec. VI A).

Section VI examines the operational side — alignment drift, counter-rotating beams, the polarity reversal, and the time budget — and Sec. VII collects the conclusions. The simulation infrastructure and the spin-signal estimator, on which all quantitative statements rest, are described first in Sec. II.

II. SIMULATION METHOD

A. Tracking tools

All results are obtained with a particle-and-spin tracking code that integrates the equations of motion element by element through the fields of the lattice of Sec. I B, using a fourth-order Gauss–Legendre symplectic integrator; the spin is propagated in the same steps by the Thomas-BMT equation. Symplectic integration prevents the slow accumulation of energy and amplitude errors, which is a prerequisite for resolving precession rates at the nrad/s level over millions of turns. The genuine EDM enters the model through a switch; with the coupling set to $\eta = 1.88 \times 10^{-15}$ the code reproduces $dS_y/dt = 9.81 \times 10^{-10}$ rad/s, odd under beam reversal, in agreement with Eq. (C1) of Ref. [1]. All false-EDM results below are obtained with the EDM switch off, so that any measured out-of-plane precession is by construction a systematic effect. Quadrupole misalignments, gradient errors (β-beating), and roll angles enter the tracker directly as element parameters.

Wide parameter scans (noise Monte Carlo, β-beating sweeps) use a second, independent layer: the lattice functions, tunes, and the orbit response matrix,

$$R_{ij} = \frac{\sqrt{\beta_i \beta_j}}{2 \sin \pi Q} (KL)_j \cos\bigl(|\phi_i - \phi_j| - \pi Q\bigr), \tag{3}$$

evaluated in closed form from the quadrupole and drift transfer matrices. This analytic layer runs in seconds, agrees with the tracker at the percent level in the vertical plane, and shares nothing with it except the configuration file, so that agreement between the two cannot arise from a common implementation error. Two of its limitations play a role later and are recorded here: it treats the electric deflectors as field-free drifts, which is accurate vertically but not horizontally (Sec. V); and an idealized response matrix can misrepresent the least-visible modes of the real beam dynamics, so response matrices used in inverse problems are always rebuilt from tracking.

B. Measuring the false EDM

The quantity of interest is the secular slope of $S_y(t)$. Extracting it reliably requires care, because the betatron oscillation of the tracked particle feeds oscillatory components into the spin motion that exceed the sought slope by orders of magnitude. Two measures are combined.

First, the tracked particle is placed on the closed orbit, where no betatron oscillation is present. The closed orbit is the fixed point of the one-turn map M in the transverse phase space (x, x', y, y'); it is found by Newton iteration, i.e. by solving $M(v) = v$ through repeated linearized steps $v \to v - (M' - I)^{-1}[M(v) - v]$, where M' is the Jacobian of the map obtained numerically from the tracker. A few iterations suffice.

Second, the spin history is fit with the model $S_y(t) = a + bt + \sum_k [c_k \cos \omega_k t + d_k \sin \omega_k t]$, and only the secular slope b is retained; residual oscillatory content is absorbed by the harmonic terms rather than biasing the slope, as it would in a plain polynomial fit.

The estimator was validated in three independent ways: (i) scanning $\sigma = 2.5$–10 μm yields $|f| \propto \sigma^p$ with $p = 2.00 \pm 0.01$, the pure quadratic scaling of a geometric phase with no linear leakage; specifically, at $\sigma = 5$ μm after orbit correction we measure $f = 0.62 \times 10^{-9}$ rad/s (median of 3 random seeds) compared to 2.72×10^{-9} rad/s at $\sigma = 10$ μm, giving an observed ratio of 4.36 in agreement with the σ^2 prediction of 4.0; (ii) under sign flips of the misalignment pattern the measured f obeys $f(+dx, -dy) = -f(+dx, +dy)$ and $f(-dx, -dy) = +f(+dx, +dy)$ at the 10^{-3} level, confirming the bilinear character of Eq. (2); (iii) the average of four particles launched symmetrically around the closed orbit reproduces the single on-orbit result within 1%.[1]

III. WHY ORBIT CORRECTION ALONE IS NOT SUFFICIENT

A. The residual after standard correction

For random misalignments of $\sigma = 10$ μm in both planes, the raw false EDM is of order 10^3 times the target [Eq. (1)]; a representative pattern used throughout this paper gives 356×. Standard orbit correction — flattening the closed orbit with steering correctors — reduces the ensemble median by a factor of about eight; combined with the counter-rotating–beam difference discussed in Sec. VI, the residual is about 62× the target ($\approx 6 \times 10^{-28}$ $e$cm), scaling as σ^2 (Fig. 1). Orbit correction is thus effective but not sufficient, and the remainder of this section identifies what the residual consists of.

B. Symmetric and antisymmetric misalignments

Let the 48 offsets of one plane form a vector v, with QF at index $2k$ and QD at index $2k{+}1$ in cell k. Within one cell the two offsets can be separated into the part in which the quadrupoles move *together* and the part in which they move *oppositely* — a simple 2×2 operation on the pair,

$$\begin{pmatrix} v^{\mathrm{s}}[2k] \\ v^{\mathrm{s}}[2k{+}1] \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} v[2k] \\ v[2k{+}1] \end{pmatrix},$$
$$\begin{pmatrix} v^{\mathrm{a}}[2k] \\ v^{\mathrm{a}}[2k{+}1] \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} v[2k] \\ v[2k{+}1] \end{pmatrix}. \tag{4}$$

The first matrix replaces both offsets by their average $\frac{1}{2}\bigl(v[2k] + v[2k{+}1]\bigr)$, giving the symmetric part v^{s}; the second replaces them by $\pm$ half their difference $\frac{1}{2}\bigl(v[2k] - v[2k{+}1]\bigr)$, giving the antisymmetric part v^{a}. The full projectors P_{sym} and P_{anti} simply apply this same 2×2 block to each of the 24 cells in turn (so $v^{\mathrm{s}} = P_{\mathrm{sym}} v$ and $v^{\mathrm{a}} = P_{\mathrm{anti}} v$); they are complementary, $P_{\mathrm{sym}} + P_{\mathrm{anti}} = I$. The same pair of matrices is applied independently to the horizontal and to the vertical offset vectors.

Carrying this out for all 24 cells re-expresses the 48 offsets of one plane as 24 symmetric numbers $m_k =$

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[1] The symmetric four-particle average must be constructed around the closed orbit; constructed around the ideal axis it retains the common betatron component induced by the orbit offset itself and fails by orders of magnitude.

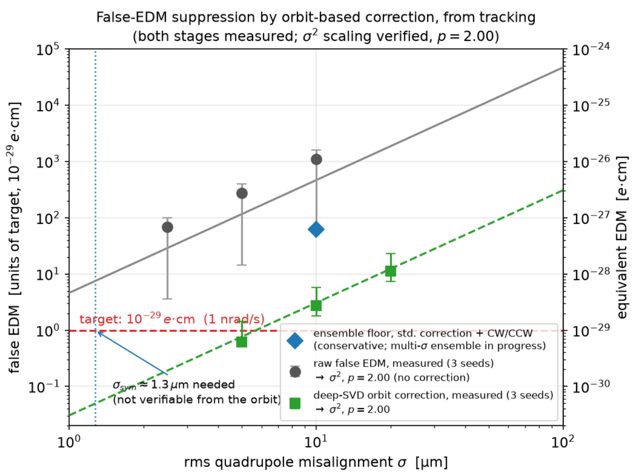

FIG. 1. False EDM before and after orbit-based correction versus the rms quadrupole misalignment, from tracking. Both stages are directly measured points, not drawn curves. Grey circles: the raw false EDM (no correction, median and full spread of three random seeds at $\sigma = 2.5, 5, 10$ μm). Green squares: the residual after a deep truncated-SVD orbit correction (three seeds at $\sigma = 5, 10, 20$ μm). The lines through each are power-law fits $|f| \propto \sigma^p$ with $p = 2.00$, confirming the quadratic scaling empirically over a factor of four in σ at both stages. The blue diamond marks the conservative ensemble floor of a standard-depth correction with the counter-rotating–beam difference at $\sigma = 10$ μm (Table II); the deep-SVD points reach lower, so they bound the slope while the conservative floor bounds the absolute level, and a multi-σ ensemble at the standard depth is in progress. The large vertical spread of the measured points is physical: at fixed σ the false EDM varies by more than an order of magnitude with the symmetric/antisymmetric content of the pattern (Sec. III C), so the rms alone does not determine the signal.

$\frac{1}{2}\big(v[2k] + v[2k{+}1]\big)$ and 24 antisymmetric numbers $d_k = \frac{1}{2}\big(v[2k] - v[2k{+}1]\big)$. Because the originals are recovered by $v[2k] = m_k + d_k$ and $v[2k{+}1] = m_k - d_k$, this is a reversible relabeling that loses nothing $(24 + 24 = 48)$: the symmetric pattern $v^{\rm s} = P_{\rm sym}v$ collects the m_k (all $d_k = 0$), the antisymmetric pattern $v^{\rm a} = P_{\rm anti}v$ collects the d_k (all $m_k = 0$), and every misalignment vector is the sum of exactly one of each, $v = v^{\rm s} + v^{\rm a}$. The two families of patterns are orthogonal and each spans half of the 48-dimensional misalignment space, so they form a complete pair of bases into which *any* pattern resolves; applied separately to the two planes this gives $v_x = v_x^{\rm s} + v_x^{\rm a}$ and $v_y = v_y^{\rm s} + v_y^{\rm a}$, the representation used in the channel decomposition of Sec. III C.

The physical distinction between the two components follows from the opposite-sign gradients of QF and QD. A misaligned quadrupole deflects the beam by an angle proportional to (gradient)×(offset). When two neighbors are displaced *oppositely* (antisymmetric component), the opposite gradients and opposite offsets combine to kicks of the *same* sign: the kicks add coherently around the ring into a slowly varying deflection pattern whose dominant azimuthal harmonics lie near the betatron tune, and the closed-orbit response is large. When two neighbors are displaced *together* (symmetric component), the kicks are opposite and cancel within each cell; what survives is a cell-to-cell alternation at azimuthal harmonic $k \approx 24$ (one sign change per cell, 24 cells). The closed-orbit response to a deflection pattern of harmonic k follows the gain law

$$G_k = \frac{C}{|Q^2 - k^2|}, \qquad C \approx 24.8,\ Q^2 \approx 5, \tag{5}$$

which is resonant for k near the tune and strongly suppressed far above it; with $Q \approx 2.3 \ll 24$ the alternating deflection pattern of a symmetric component is nearly invisible [Fig. 2]. Quantitatively, a 10-μm-rms random pattern leaves an rms orbit trace of ~ 100 μm if antisymmetric but ~ 19 μm if symmetric (tracking-derived response), and the two traces differ in kind, not only in size: the symmetric trace is almost entirely *leakage*, a few-percent admixture of the pattern into the strongly-coupled modes. Orbit correction removes exactly that visible slice, so it erases most of an antisymmetric pattern (rms reduced by a factor $\sim$3 in our simulations) but leaves a symmetric pattern nearly intact (rms reduced by only $\sim$20%).

The same conclusion can be read off the response matrix itself without reference to harmonics. The singular-value decomposition of R supplies 48 mutually orthogonal, unit-normalized misalignment patterns v_i, each with its own orbit strength σ_i (pattern v_i produces an orbit of size σ_i). We ask how symmetric each of these patterns is by applying to it the very same projector $P_{\rm sym}$ of Eq. (4): the *symmetric fraction*

$$s_i = \|P_{\rm sym}\, v_i\|^2 \in [0, 1] \tag{6}$$

is the squared length of the pattern's projection onto the symmetric subspace, so that $s_i = 0$ for a purely antisymmetric pattern and $s_i = 1$ for a purely symmetric one. Equation (6) thus turns the decomposition of Eq. (4) into a single number per SVD mode that can be plotted against its orbit strength. Plotting σ_i against s_i [Fig. 2, left] shows the strongly-coupled patterns to be almost purely antisymmetric ($s_i \approx 0$) and the weakly-coupled ones almost purely symmetric ($s_i \approx 1$); the overall visibility ratio between the strongest and weakest patterns is $\mathrm{cond}(R) = 193$. Orbit correction, which by construction removes what the orbit sees, therefore removes the antisymmetric component and leaves the symmetric one.

C. The false EDM as a bilinear form: four channels

The connection between this decomposition and the false EDM follows directly from the physics of Sec. I A, with no further formalism. A horizontally misaligned quadrupole precesses the spin about the vertical axis by a small angle proportional to its offset $v_{x,i}$; a vertically

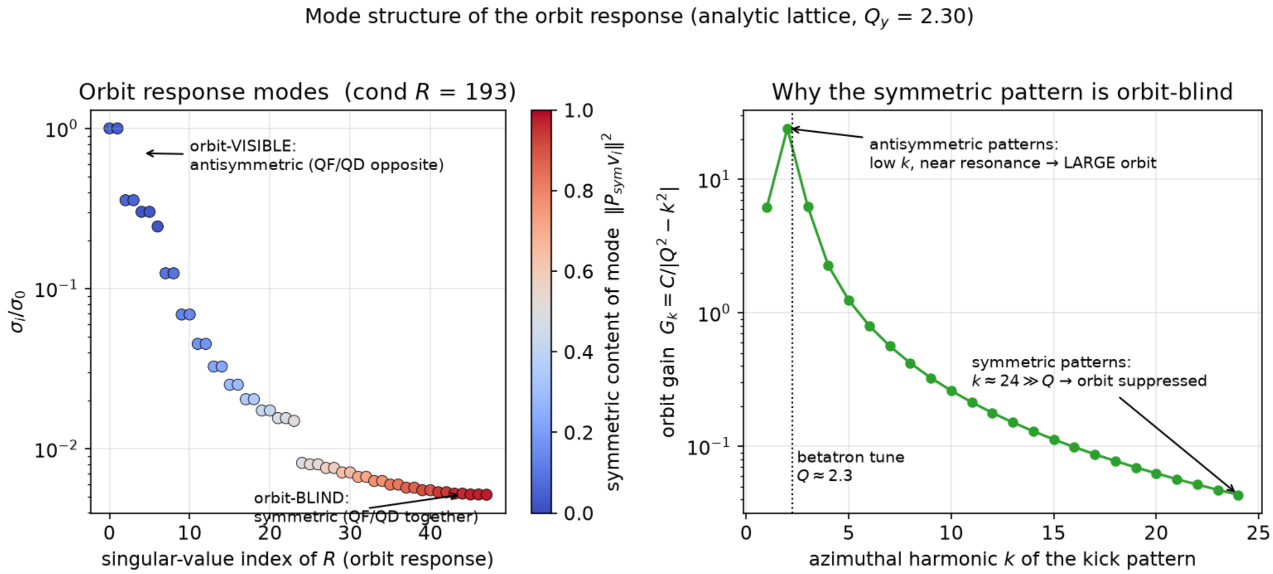

FIG. 2. Orbit visibility of misalignment patterns (analytic lattice, $Q_y = 2.30$). Left: independent misalignment patterns ranked by the strength of their orbit response; patterns that couple strongly to the orbit are antisymmetric (QF, QD opposite), patterns that couple weakly are symmetric (QF, QD together); the visibility ratio is cond$(R) = 193$. Right: the orbit gain law of Eq. (5); antisymmetric patterns drive low harmonics near the betatron resonance, symmetric patterns drive $k \approx 24$, far above it.

misaligned one precesses it about the radial axis by an angle proportional to $v_{y,j}$. Two precessions about different axes do not commute, and the out-of-plane residue left by each ordered pair of them is proportional to the *product* of the two small angles. Summing this over all pairs of quadrupoles around the ring,

$$f \;=\; \sum_i \sum_j W_{ij}\; v_{x,i}\, v_{y,j} \;\equiv\; B(v_x, v_y), \tag{7}$$

where the pairwise weights W_{ij} depend only on the lattice — on where the two magnets sit and on how the orbit deflection started by one is carried around the ring to the other [the orbit response, Eq. (3)] — and not on the misalignments themselves. Equation (7) states that the false EDM is *bilinear*: every term contains exactly one horizontal and one vertical offset, never two of the same plane. This is the sign structure verified directly in Sec. II B.

Splitting each plane independently into its symmetric and antisymmetric parts, $v_x = v_x^{\mathrm{s}} + v_x^{\mathrm{a}}$ and $v_y = v_y^{\mathrm{s}} + v_y^{\mathrm{a}}$ [Eq. (4)], and expanding the double sum, the terms fall into four groups:

$$f = \underbrace{B(v_x^{\mathrm{s}}, v_y^{\mathrm{s}})}_{f_{ss}} + \underbrace{B(v_x^{\mathrm{s}}, v_y^{\mathrm{a}})}_{f_{sa}} + \underbrace{B(v_x^{\mathrm{a}}, v_y^{\mathrm{s}})}_{f_{as}} + \underbrace{B(v_x^{\mathrm{a}}, v_y^{\mathrm{a}})}_{f_{aa}}. \tag{8}$$

The two planes are mixed by the weights W_{ij}, not by the projectors. Because a misalignment component acts on the spin mainly through the orbit it drives, and that orbit is large for an antisymmetric component and G_k-suppressed for a symmetric one [Eq. (5)], the pure-symmetric channel f_{ss} is by far the smallest, and every channel containing an antisymmetric factor is larger.

We stress that f_{sa} and f_{as} are two distinct physical quantities, not a double counting. In f_{sa} the *horizontal* pattern is symmetric and the *vertical* one antisymmetric; in f_{as} the roles are exchanged. Nothing forces these to be equal: the two planes enter the ring differently (only the horizontal plane carries the bending), and the weights W_{ij} are not symmetric under exchanging the planes, because the order of two non-commuting precessions matters. The tracker confirms the difference directly: for the representative pattern, $f_{as} = -272\times$ target while $f_{sa} = +28\times$ — an order of magnitude apart and of opposite sign.

Each channel is a separate tracking measurement, not a fit, and this measurement underlies all symmetric/antisymmetric attributions in this paper. The misalignment pattern (v_x, v_y) is projected once with Eq. (4), and the tracker is then run four times. In each run the quadrupole offsets of the machine are *set to* one projected part per plane — for f_{as}, for example, the horizontal offsets are the pattern v_x^{a} and the vertical offsets are v_y^{s}, with no other imperfections — and the estimator of Sec. II B returns that channel's f directly. The weights W_{ij} never need to be known or computed: they are properties of the lattice that the tracker itself embodies, and Eq. (7) enters only through its structural consequence that every term contains exactly one horizontal and one vertical offset, so that the four runs isolate the four groups of terms. Bilinearity is then checked rather than assumed: Figure 3 shows the result at $\sigma = 10\ \mu$m: the four signed channels sum to the full-pattern false EDM within

0.1% ($f/\sum = 0.999$–1.001 over three seeds), confirming Eq. (8); the pure-symmetric channel is the smallest in every seed; and each channel individually scales as σ^2. The measured channel ratios are consistent with the bilinear structure: one orbit-trace factor of ~ 5 enters per plane, so f_{aa}/f_{ss} of order tens is expected and observed. The measured channels are used as the quantitative reference throughout.

Two consequences frame the rest of the paper. First, the false EDM is not a function of the rms misalignment: it is σ^2-homogeneous, but at fixed σ its value depends on the pattern — in Fig. 3 the channel magnitudes at the same 10 μm rms range from $\sim$20× to $\sim$400× the target in the three-seed means, and from $\sim$1× to $\sim$900× across individual seeds. A single alignment tolerance therefore cannot summarize the systematic, and a correction procedure changes not only the size but the *statistics* of the misalignment: it converts a random pattern into a structured one, concentrated in whichever component the procedure cannot see. Second, the division of labor for suppression is fixed by Eq. (8): orbit correction removes the antisymmetric factors and with them the three large channels; the remaining floor is f_{ss}, and any further progress requires a measurement that reaches the symmetric component itself.

IV. THE SYMMETRIC COMPONENT CANNOT BE RECOVERED BY INVERSION OR AMPLITUDE READOUT

Before presenting the measurement that succeeds, we summarize the techniques that do not, and why. All of them attempt to reconstruct the misalignments (including, crucially, their symmetric part) from orbit readings. The realistic error model includes static BPM offsets ($\sim$100 μm each, the electronic zero of a monitor relative to the true axis), white BPM noise ($\sim$1 μm per reading, reducible by averaging), and a coherent optics-model error (β-beating, implemented as per-quadrupole gradient errors of 0.5–5%; a machine calibrated with standard LOCO techniques typically retains $\sim$1%).

Difference-orbit inversion. Modulating all quadrupole gradients between two settings and subtracting the orbits cancels the static BPM offsets exactly and leaves a linear system $\Delta y = \Delta R\, \Delta q$. With no errors the inversion is exact. Its weakness is conditioning: $\mathrm{cond}(\Delta R) = 3.7 \times 10^4$, with the weak directions being precisely the symmetric ones (established by the same symmetric-fraction analysis as in Fig. 2, applied to ΔR), so reconstruction errors are amplified along the component of interest. Regularizing the inversion — truncated SVD or Tikhonov damping — does not help here: truncation discards precisely the symmetric directions, trading their noise amplification for a full bias of the sought component. White noise can be beaten by synchronous detection — modulating at $\sim$ 1 kHz and averaging reaches an effective noise of $\sim$ 14 nm within minutes — but the optics-model error cannot: it is coherent, identical in every average, and at β-beating of only 0.5% the reconstructed symmetric component is several times larger than the true one (283 μm against a 41-μm signal in our standard test) while the antisymmetric component remains accurate [Fig. 4]. A caution for the evaluation of such reconstructions: the Pearson correlation between reconstruction and truth, dominated by the well-reconstructed antisymmetric components, remains as high as 0.7–0.99 in these failures; the meaningful metric is the rms error of each component separately, compared with the size of that component in the signal. Because the limit is coherent, it is independent of BPM technology; lower-noise pickups, including SQUID-based ones, shorten the integration time but do not move the floor. The same structural point can be stated as a bound: any estimator that cancels static offsets by a two-setting difference pays for the cancellation with a $\|\Delta R^{-1}\| \sim \|R^{-1}\|/\varepsilon_g$ amplification, where ε_g is the relative setting difference.

Per-quadrupole modulation read as an amplitude. Assigning each quadrupole its own modulation frequency and reading the per-frequency amplitude at the BPMs — the ac-BBA concept of Ref. [5] — promises a direct, inversion-free readout: a quadrupole whose center is offset from the beam by o_i imprints, when its gradient is modulated, a dipole-like deflection proportional to o_i (the *feed-down* kick, the dipole field component seen by a beam passing off-center). The method fails in this ring for an instructive reason [Fig. 5]. The problem is caused by the modulated quadrupole itself, acting globally. A quadrupole is a focusing element: changing its gradient by even 2% slightly shifts the tune and the focusing functions of the *entire* ring. The pre-existing closed orbit — built by the offsets of all the other quadrupoles, and at $\sim$0.37 mm rms several hundred times larger than the $\sim$ 0.9-μm feed-down signal being sought — is retransported through this slightly altered focusing and moves by a few μm at every BPM. This global response, which we call optics *breathing*, appears at exactly the modulation frequency of the quadrupole being measured, so no frequency filter can separate it from the signal; it exceeds the signal about sevenfold. A decomposition test isolates the two contributions cleanly: with the full misalignment pattern present, modulating one quadrupole moves a BPM reading by -9.83 μm; deleting every offset except that quadrupole's own leaves only -0.037 μm. The demodulated amplitude is therefore reading the orbit built by the *neighbors*, 266 times more strongly than the offset it was meant to measure, and dividing it by a calibration cannot repair a number that measures the wrong quantity. Because breathing is coherent with the modulation, it does not average away with more BPMs or lower noise; removing it in an idealized model restores perfect reconstruction, which pins the failure entirely on this mechanism.

Learned and structured estimators. Because the misalignment-to-orbit map is linear, a trained neural network can at best relearn the (regularized) inverse of R;

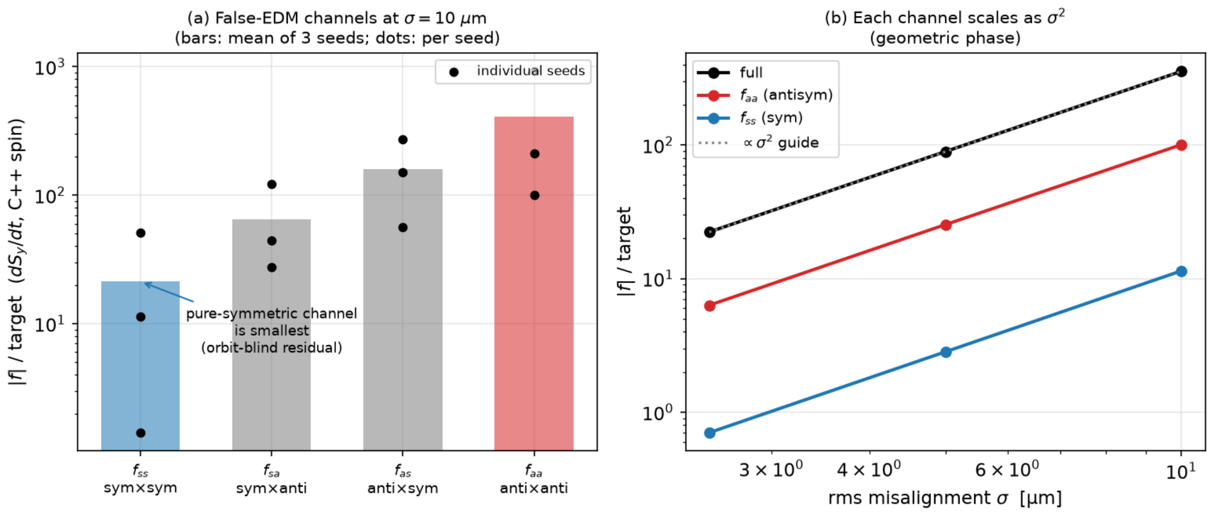

FIG. 3. The four false-EDM channels of Eq. (8), measured by projecting the misalignments and tracking each part separately. (a) At $\sigma = 10$ μm the pure-symmetric channel f_{ss} is the smallest, and every channel containing an antisymmetric factor is larger (bars: mean of three seeds; dots: individual seeds; the spread reflects the signed, bilinear character of f). (b) Each channel scales as σ^2. The four signed channels sum to the full signal within 0.1%.

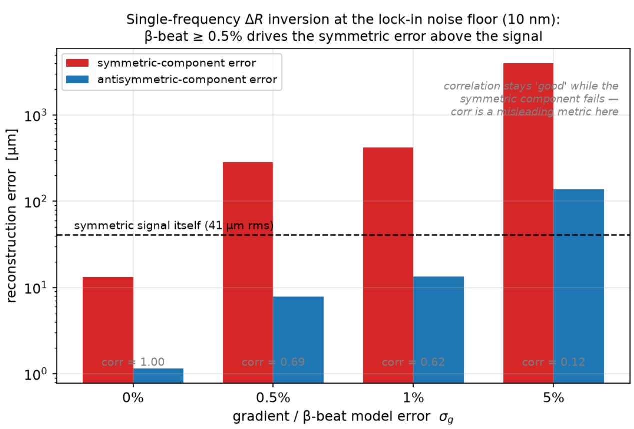

FIG. 4. Difference-orbit inversion at the synchronous-detection noise floor (10 nm, 40 seeds): reconstruction error of the symmetric and antisymmetric components versus the optics-model error. The dashed line marks the size of the symmetric signal itself. A model error at the realistic $\geq 0.5\%$ level drives the symmetric error far above the signal while the Pearson correlation (gray labels) remains high.

in our tests it matched truncated-SVD inversion on both components without improving either (symmetric error 5.6 vs. 6.3 μm against a 9.9-μm signal) — the limit is the information in the data, not the flexibility of the estimator. Sparsity priors (LASSO, CLEAN, greedy harmonic selection) reach the same floor, and harmonic fits of a known signature trade the conditioning problem for a bias problem (one harmonic coefficient lumps together field, misalignment, and BPM-offset content). A calibrate-then-monitor strategy (LOCO followed by drift monitoring) tracks *changes* of the antisymmetric component well (6.5 μm rms under 50 μm BPM offsets) but passes through the same response matrix and is equally blind to the symmetric one.

Counter-rotating beams, polarity switching, and the tune. The control chain of Ref. [1] monitors the *separation* of the two counter-rotating (CR) closed orbits and reduces it with correctors. The CR separation, however, is itself an orbit observable and inherits the orbit's visibility structure: in our tracking the antisymmetric-to-symmetric signature ratio is 5.7× in a single-beam orbit and 8.0× in the CR separation (an independent pattern set gives 3.8× and 4.5×) [Fig. 6] — the CR separation opens no additional window on the symmetric component. A related question is why the beam-reversal comparison does not cancel the geometric phase outright. In the idealized periodic lattice we measure the reversed beam to be exactly equivalent to the forward beam with all quadrupole polarities flipped (ratio 1.00); and the polarity flip is *not* a symmetry of the geometric phase, because the phase is an ordered accumulation of non-commuting precessions — reversing the traversal order is not equivalent to reversing the sign. Measured flip ratios $f(-g)/f(+g)$ of +2.8, +0.85, and +1.17 for symmetric, antisymmetric, and generic patterns confirm that the phase is approximately even in the gradient but with pattern-dependent deviations; consequently the CW/CCW difference cancels a pattern-dependent fraction of the false EDM — a factor ~ 3 in practice (Sec. VI) — rather than all of it, and polarity switching provides no independent cancellation knob for the quadrupole-alignment geometric phase (it remains effective against systematics that are odd in the gradient). Finally, raising the betatron tune toward the symmet-

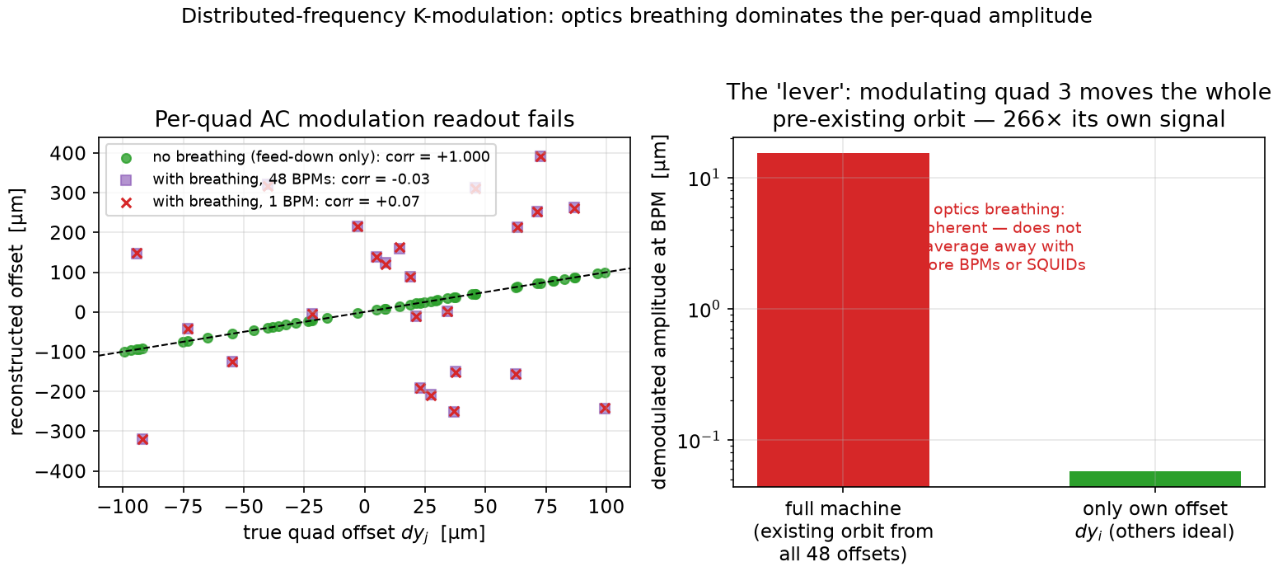

FIG. 5. Per-quadrupole gradient modulation read as an amplitude. Left: reconstructed versus true offsets — exact when the optics breathing is artificially removed (green); uncorrelated with it present, whether one BPM (red) or all 48 (purple) are used. Right: the demodulated amplitude responds to the re-transported pre-existing orbit at 266× the level of the quadrupole's own feed-down signal; the contamination is coherent and does not average down.

ric kick harmonic would make the symmetric component visible in principle; in practice the stopband limit (180° phase advance per cell) caps the tune at $Q \leq 12$, recovering only the fast-varying part of the symmetric family (its high azimuthal harmonics m, where m indexes the harmonics *within* the 24-dimensional symmetric subspace, the slowly-varying members requiring $Q \geq 16$), and the stronger gradients required ($0.21 \rightarrow 0.69$ T/m for Q_y : $2.3 \rightarrow 10.3$) raise the false EDM itself steeply ($\sim g^3$, a measured factor of 32) while the genuine EDM signal, set by the electric field, is unchanged. A further alternative outside the orbit toolbox — operating slightly off the magic momentum to sense the geometric phase resonantly — fails for reasons detailed in Appendix A.

The common conclusion of this section is structural. For every technique in the inversion and amplitude-readout families, the limiting effect is coherent — conditioning amplified by β-beating, or optics breathing — and coherent effects do not average down. *Within the simulated model, no BPM technology — including SQUID-based BPMs — changes these floors.* We emphasize the scope of this statement. It is a structural property of the estimator classes considered, established by class-level arguments (the conditioning bound and the coherence of the breathing term) together with systematic simulation; it is not an information-theoretic impossibility claim for all conceivable orbit-based measurements. Indeed, the next section exhibits an orbit-based measurement outside these classes that does reach the symmetric component.

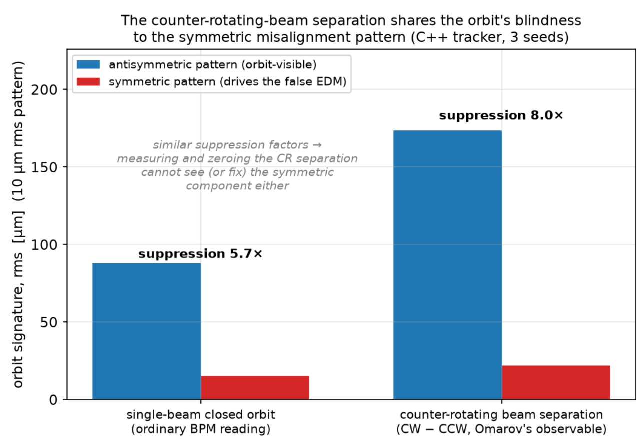

FIG. 6. The counter-rotating beam separation (right group) suppresses the symmetric pattern by a factor similar to the ordinary single-beam orbit (left group): as an orbit observable it shares the orbit's blindness to the component that drives the residual false EDM.

V. REACHING THE SYMMETRIC COMPONENT: NULL-FINDING BBA

A. Procedure

Standard quadrupole beam-based alignment does not invert a matrix and does not read an amplitude against a calibration; it locates each magnetic center as the beam position at which the response to a gradient change *vanishes*. For each of the 47 alignable quadrupoles in turn:[2] (i) using the nearest steering corrector, scaled by the single response-matrix element R_{ij}, the beam is placed at two positions bracketing the expected center of quadrupole i; (ii) at each position the gradient of quadrupole i is changed by a small amount and the difference of the two closed orbits is recorded at all 48 BPMs — this difference is the feed-down imprint, proportional to the beam–center offset; (iii) since the imprint changes sign as the beam crosses the center, the center is obtained as the weighted least-squares zero over the BPMs, with the spread of the per-BPM zero crossings retained as a per-magnet quality measure. All orbits are read with the beam placed on the 4D closed orbit (Sec. II B). In the simulation the steering corrector is a controlled transverse displacement of a quadrupole adjacent to the one under test; in the experiment the same dipole kick can be produced without moving any magnet, by a current asymmetry between the quadrupole coils, which shifts the magnetic center electrically. The two are equivalent — an offset quadrupole and an on-axis quadrupole with a superimposed dipole field act identically on both orbit and spin.

The role of the response matrix in this procedure is confined to the *forward* direction: a single, well-conditioned element sizes the steering bump. No global inverse appears anywhere, so the conditioning wall of Sec. IV does not apply; the 47 centers are measured locally and independently, and the symmetric combination of the answers is determined as well as any other combination. An error in the forward element mis-sizes a bump and biases one local measurement — an effect that will be visible below — but it does not re-introduce the collective visibility gap.

B. Idealized and realistic optics

In idealized optics the procedure works outright: the symmetric centers are recovered, and the false EDM of the representative pattern drops from 356× to 1.6× the target.

Under a realistic 1% β-beating a single pass fails, through the same breathing mechanism as in Sec. IV: modulating one quadrupole re-transports the large uncorrected orbit through the beating optics and biases the zero crossing. The remedy is iteration — correct the orbit with the current estimates, then re-measure — since a smaller orbit breathes less. Iterating with under-relaxation and the per-magnet quality cut, the vertical plane converges cleanly (residual halving each pass, reaching 0.9 μm). The horizontal plane initially stalled, and the diagnosis is instructive. The steering bumps were sized with the analytic optics model, which treats the deflectors as drifts (Sec. II A). Vertically this is accurate, and indeed the model reproduces the tracker's vertical tune to four decimal places (fractional tune 0.3032 in both). Horizontally the deflectors focus and bend, and the model does not know it: it predicts equal tunes in the two planes, while the tracker's horizontal fractional tune is higher by 0.052; element by element, the model's horizontal response matrix is essentially uncorrelated with the measured one (median ratio -0.05), and it selects the wrong steering corrector for 36 of the 47 magnets, against a 5%-accurate response and 12 wrong selections vertically. Re-building the steering from the *measured* response matrix — ordinary practice on a real machine — repairs the horizontal plane: the per-magnet quality ceases to throttle the horizontal corrections, and the false EDM after three passes improves ninefold over the analytic-model steering (Table I). The measured matrix is obtained in the simulation exactly as it would be on the machine: each quadrupole in turn is displaced by a known amount, the resulting closed-orbit change is recorded at all 48 BPMs, and the ratio (orbit change)/(displacement) fills one column of the matrix.

TABLE I. Iterated null-finding BBA under 1% β-beating (tracking, three passes), steering with the idealized analytic model versus the measured response matrix. "sym dx/dy" are the symmetric residuals of the measured-matrix run. Both runs solve the vertical plane; only the measured matrix also corrects the horizontal one.

pass	sym dy [μm]	sym dx [μm]	$\lvert f\rvert$ / target analytic	$\lvert f\rvert$ / target measured
raw	—	—	356×	356×
1	3.27	4.27	368×	73×
2	1.70	2.90	238×	**17×**
3	0.91	2.29	145×	28×

C. Closing the chain: a final orbit correction

The 17–28× residual of the measured-matrix run is not the symmetric floor. This is the central point of the whole scheme, and Fig. 7 makes it visible: across the BBA passes the symmetric residual falls steadily (the vertical component to below the target-equivalent level), yet the false EDM $|f|$ does not follow it down — it stalls near 20× and even scatters upward, because f is a signed bilinear quantity dominated by a different component of the residual. Decomposing the third-pass residual with the channel method of Sec. III C shows what that component is: the symmetric part contributes 3.5× the target, the antisymmetric part contributes 16× — a ~ 2 μm residual centering error at BBA's per-magnet precision floor, largest in the horizontal plane, which stays the harder

[2] The gradient-modulated quadrupole of cell 0 is a distinct element type in our model and is excluded; its offset is set to zero and reported as such.

one to center even after the measured-matrix steering of Sec. V B has brought this channel down from $\sim 145\times$ (analytic steering) to $\sim 16\times$ — and cross terms account for the remainder, in the ordering expected from Eq. (8). The residue is thus the ordinary precision floor of the centering rather than the analytic-steering failure, which the measured matrix has already cleared. We emphasize that BBA itself has no preference between the two subspaces: it measures each magnet independently, so its errors are uncorrelated from magnet to magnet and land with equal weight in the symmetric and antisymmetric components — it lowers *both* to the same few-μm floor. The asymmetry visible in Fig. 7 is one of leverage, not of cleaning: at that common floor the antisymmetric part still dominates f through its stronger channels [Eq. (8)], while the symmetric part is already near its target-equivalent level. So BBA does the job only it can do — reaching the orbit-blind symmetric part — but leaves an antisymmetric residue at its per-magnet precision floor that more passes do not remove. That leftover is orbit-visible, and a single standard orbit correction removes it: applied to the BBA residual, it brings the single-beam false EDM from $28\times$ to 0.03–$0.8\times$ the target, depending on the correction depth. Scaling the corrected residual down confirms a clean σ^2 dependence with no additive floor over the tested range.

The resulting picture is consistent and complementary: *orbit correction removes the antisymmetric channels of Eq. (8); null-finding BBA reduces the symmetric channel that orbit correction cannot see; in combination they reach the target, while each alone stalls* ($62\times$ for orbit correction, $\sim 20\times$ for BBA without the final correction). The division of labor is also one of *precision*: each per-magnet zero crossing carries a $\sim \mu$m-level measurement floor (residual breathing and steering-model error), so iterating BBA alone approaches that floor only slowly; the antisymmetric residue, by contrast, needs no per-magnet measurement at all — the orbit displays it directly and resonantly amplified, and nulling it collectively with a truncated-SVD correction (the "depth" above is the singular-value cutoff) is far more precise than measuring it magnet by magnet. Table II summarizes the chain. Two caveats bound the claim. The sub-target endpoint is a single misalignment seed of a signed, bilinear quantity whose value scatters from seed to seed (Fig. 3); the robust statement is suppression to the symmetric floor of a few times the target, and a multi-seed campaign to fix the distribution is in progress. And the entire analysis assumes that gradient modulation does not itself move the magnetic centers. The air-core (iron-free) quadrupoles foreseen for the experiment eliminate magnetic hysteresis, but thermal or mechanical motion of the coil geometry at the micron level would enter the measurement directly, and this hardware floor has not been modeled.

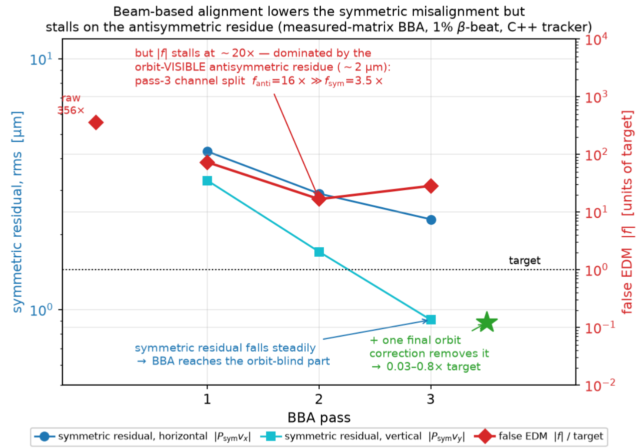

FIG. 7. Beam-based alignment lowers the symmetric misalignment but stalls on the antisymmetric residue (measured-matrix BBA under 1% β-beating, C++ tracking). Blue/cyan (left axis): the symmetric residual of the two planes falls steadily with each pass, the vertical one below the target-equivalent level — BBA reaches the orbit-blind component that response-matrix inversion cannot. Red (right axis): the false EDM $|f|$ nonetheless stalls near $20\times$ the target, because it is dominated by the orbit-visible antisymmetric residue ($\sim 2\ \mu$m), the centering error remaining at BBA's precision floor (largest in the horizontal plane) once the measured-matrix steering of Sec. V B has done its work; the channel decomposition of the third-pass residual gives $f_{\mathrm{anti}} = 16\times \gg f_{\mathrm{sym}} = 3.5\times$. A single final orbit correction removes that residue (green star), bringing $|f|$ below the target for this seed.

TABLE II. The suppression chain at $\sigma = 10\ \mu$m (single beam, tracking). The second row is an ensemble median including the counter-rotating–beam difference; the remaining rows follow the representative pattern (raw $356\times$). The equivalent EDM uses 1 nrad/s $\leftrightarrow 10^{-29}$ $e$cm. The final entry is a single-seed result; the multi-seed distribution is under study.

Stage	$\lvert f\rvert$ / target	EDM [$e$cm]
raw	$356\times$	$\sim 4\times 10^{-27}$
orbit corr. + CW/CCW	$\sim 62\times$	$\sim 6\times 10^{-28}$
BBA (measured steering)	17–$28\times$	$\sim 2\times 10^{-28}$
BBA + final orbit corr.	0.03–$0.8\times$	$\lesssim 10^{-29}$

VI. OPERATING THE SCHEME: DRIFT, COUNTER-ROTATING BEAMS, AND THE TIME BUDGET

An alignment campaign is useful only if its result survives on the machine. Quadrupole positions drift — thermally and with ground motion — so the corrected state of Sec. V is a snapshot, onto which the machine continuously re-loads random misalignment. We simulated this directly by adding fresh random offsets of rms σ_{d} to the corrected state and re-measuring the false EDM, with and without orbit feedback.

Without any re-correction the false EDM returns quickly: $\sigma_{\mathrm{d}} = 10$ μm restores hundreds of times the target. With *continuous orbit correction* — routine, fast, and non-invasive on any storage ring — the picture changes qualitatively, because random drift is shared between the two components and the orbit feedback continuously removes its antisymmetric part: the false EDM stays below the target for σ_{d} up to about 5 μm (0.05× at 2 μm, 0.3–0.6× at 5 μm) and reaches only a few times the target at 10 μm. Only the slowly accumulating *symmetric* part of the drift requires a repetition of the BBA campaign. The alignment therefore does not need to be *held* at the micron level mechanically; it needs to be re-established at a cadence set by the symmetric drift rate.

A. Where orbit correction substitutes for the polarity reversal

The systematic-error program of Ref. [1] pairs beam-based alignment with a signed combination of measurements: the genuine EDM is odd under beam reversal, so it survives the counter-rotating difference $(dS_y/dt)_{\mathrm{CW}} - (dS_y/dt)_{\mathrm{CCW}}$, while common-mode systematics cancel [Eq. (C1) of Ref. [1]]. Two beams stored simultaneously make this difference drift-free. A false EDM that were purely beam-reversal-even would vanish here; the geometric phase is not, because the counter-rotating beams ride slightly different closed orbits, and a beam-reversal-*odd* remnant survives. In our tracker this remnant, at $\sigma = 10$ μm, is 14× the target for a purely symmetric misalignment and 491× for a purely antisymmetric one. Orbit correction removes the antisymmetric part; the symmetric remnant is the same orbit-blind floor as before, now reached only by BBA.

Reference [1] removes even this remnant with a second knob: *reversing the polarity of all focusing quadrupoles* and repeating the counter-rotating difference [its Eq. (C2)]. Under simultaneous reversal of the beam direction and the magnetic gradient the Lorentz force $q\,\boldsymbol{v} \times \boldsymbol{B}$ is unchanged, so a counter-rotating beam in the reversed lattice retraces the trajectory of a co-rotating beam in the nominal one; the magnetic-lattice false EDM is therefore identical between the two configurations and cancels in the four-fold combination, whereas the electric-field EDM signal, untouched by the polarity reversal, does not. We confirm this directly: the four-fold combination removes the false EDM of *both* the symmetric and the antisymmetric patterns to the estimator floor (0.009× and −0.017× the target), independently of quadrupole roll or β-beating, which respect the same $\boldsymbol{v} \times \boldsymbol{B}$ symmetry.

The four-fold cancellation is thus exact in principle, but it is not free, and it does not displace the alignment. The two polarity states cannot be stored at once; they are measured in sequence, and the cancellation requires the machine to present the *same* misalignment in both. Any change of alignment between them re-opens the false EDM, and the sensitivity is steep: a 1 μm symmetric misalignment drift between the two polarity states produces a residual of tens of times the target in our tracker. Crucially, this residual scales with the ambient alignment level — the same 1 μm drift gives 5× the target on an orbit-corrected machine (symmetric rms ~ 5 μm) but only 0.07× after BBA (symmetric rms ~ 2 μm). The drift that matters is the symmetric one, invisible to the orbit and therefore outside the reach of the continuous orbit feedback that holds the antisymmetric alignment; only a BBA-level symmetric alignment, held by periodic re-alignment, keeps the polarity-reversal cancellation intact. The air-core quadrupoles foreseen for the experiment make the electrical reversal clean and reproducible, but the change in the inter-coil Lorentz forces upon reversal is a direct source of exactly this inter-configuration displacement.

Two conclusions follow, and they set the role of orbit correction in the alignment program. First, orbit correction cannot replace BBA: the symmetric component is invisible to it in every configuration. Second, added to BBA, orbit correction can replace the polarity reversal. It removes the antisymmetric channel — the part the polarity difference would otherwise cancel — continuously, non-invasively, and without a second pass of the experiment, while BBA reaches the symmetric channel that neither orbit correction nor a single polarity state resolves. In our tracking the drift-free counter-rotating difference after BBA and orbit correction reaches 1.6× the target for the representative seed, at the same level the polarity reversal targets, without doubling the beam time and without exposing the measurement to the inter-configuration displacement. Should the polarity reversal be used in addition, the analysis above shows it still requires the BBA-level symmetric alignment to deliver its cancellation — so BBA is required on either path, and orbit correction is what stands in for the polarity reversal. An ensemble determination of the residual on both paths is left for the multi-seed campaign.

These sensitivities translate into a time budget whose absolute numbers require one machine-specific input that simulation cannot supply: the drift rate. The arithmetic is simple enough to display. One BBA pass consists of 47 magnets × 2 planes × 2 beam positions × 2 gradient settings ≈ 400 orbit acquisitions; at ~ 10 s per acquisition (orbit settling plus averaging) a pass costs about an hour of beam time, and the three-pass iteration with its intervening orbit corrections a few hours — of order half a day with overheads. Continuous orbit correction runs during data taking at no extra cost. The re-alignment cadence follows from the drift rate: if the total random drift is of order 1 μm/day — a representative figure for a thermally stabilized installation, to be confirmed for the actual site — its symmetric part ($\approx 1/\sqrt{2}$ of it, i.e. $\sim$0.7 μm/day) reaches the 2 μm scale in about three days. A few-hour BBA campaign every ~ 3 days is a duty-cycle loss at the percent level. The dominant time scale of the experiment remains the statistical one: polarimetry reaches the nrad/s level only over months to a year of running [10], and the alignment maintenance described here fits inside that schedule provided the drift rate is not far above the representative figure. The unmodeled hardware floor — magnetic-center motion under gradient modulation — enters this budget as a repeatability limit of each BBA campaign and is the main open item on the operational side.

VII. CONCLUSIONS

The quadrupole-misalignment false EDM of the symmetric-hybrid proton EDM ring is controlled by the interplay of two misalignment components with sharply different orbit visibility. The central conceptual result of this work is the bilinear four-channel structure of Eq. (8), which explains both why standard orbit correction stalls and which measurement completes it; the quantitative picture is as follows.

(1) Standard orbit-based control — orbit flattening together with the counter-rotating–beam difference — suppresses the false EDM by roughly an order of magnitude and no further: it removes the orbit-visible, antisymmetric component of the misalignment, and the weakly observable symmetric component sets a floor of $\sim 6 \times 10^{-28}$ $e$cm at $\sigma = 10$ μm, scaling as σ^2.

(2) The false EDM is a bilinear functional of the two transverse misalignment patterns [Eq. (8)], verified in tracking at the 0.1% level. It is therefore not characterized by an rms tolerance: at fixed σ it varies by more than an order of magnitude with the pattern. More generally, a correction procedure changes the statistics of the misalignment — it converts random errors into a structured residual concentrated in the component the procedure cannot see — so the driving question is not how small the residual is, but in which component it lies.

(3) In our model, the symmetric component is unreachable from orbit readings by the inversion and amplitude-readout families of techniques, for coherent reasons (conditioning amplified by optics-model error; optics breathing) that improved BPM technology does not mitigate. The counter-rotating beam separation, being an orbit observable, shares this blindness; polarity switching provides no independent cancellation for the quadrupole-alignment geometric phase; and tune elevation is capped by the stopband well below the symmetric kick harmonic.

(4) Null-finding BBA is not subject to these limits, because it locates each magnetic center by a local zero crossing rather than through a global inverse. It requires a faithful *horizontal* steering model — in this lattice the electric deflectors focus horizontally, and an idealized model that neglects this selects wrong correctors and stalls the horizontal plane — and it leaves an antisymmetric residue that a final orbit correction removes. The full chain, BBA with measured-matrix steering plus orbit correction, suppresses the single-beam false EDM from $356\times$ to the target level in our simulations (single seed; multi-seed campaign in progress), with the CW/CCW difference contributing a further factor of 2–3. Two qualifications belong to this statement. The ninefold gain of the measured over the idealized steering matrix is a property of the fidelity of the optics model supplied to the procedure, not of the algorithm, which is standard; and the chain assumes throughout that the magnetic centers remain stationary under the gradient modulation itself.

(5) Operationally, the corrected state need not be held statically: continuous orbit correction absorbs random drift up to ~ 5 μm rms, and only the symmetric drift component requires periodic re-alignment, at a cadence compatible with the statistical time scale of the experiment for drift rates of order 1 μm/day.

The paper thus both diagnoses the limitation of orbit-based control and demonstrates its resolution at the proof-of-principle level; it is not yet a tolerance document. The principal open items are the magnetic-center stability under gradient modulation (the hardware floor of the BBA measurement, not yet modeled), the multi-seed distribution of the final residual, the ensemble determination of the CW/CCW factor, and the verification of the quantitative results on other lattice variants. All numbers refer to the symmetric-hybrid ring designed for the proton EDM experiment [1], in a linear model without sextupoles; the structural conclusions — the visibility split, the bilinear channel decomposition, and the complementarity of orbit correction and BBA — follow from the alternating-gradient principle on which that ring, like any strong-focusing machine, is built.

ACKNOWLEDGMENTS

Appendix A: Off-momentum spin sensing

For completeness we document an attempted escape outside the orbit toolbox: sensing the geometric phase with the spin itself by storing the beam slightly off the magic momentum. Away from the magic momentum the in-plane spin precession unfreezes and the spin precesses at a tune $\nu_s \neq 0$; the geometric-phase contribution to the vertical spin component then appears as an oscillation whose amplitude is *enhanced* by $1/\nu_s$, suggesting a fast, resonant readout of the misalignment channel that could bypass the orbit entirely.

The idea fails against a first-order background. Misalignments not only generate the second-order geometric phase; they also tilt the stable spin direction (the invariant spin axis) at *first* order in σ. Off-magic, this tilt produces a vertical spin oscillation at the same frequency ν_s as the sought signal and exceeds it, at $\sigma = 10$ μm, by a factor of order 3600. We tested the available separation channels and found all closed: the two terms scale indistinguishably with the momentum offset; their relative phase is pattern-dependent (spread of 91° over seeds), so no fixed phase gate isolates the signal; symmetric four-particle sampling does not cancel the first-order term, which is common to the four particles; the amplitude crossing between the two terms lies at an unphysical $\sigma \approx 62$ mm; a full three-dimensional fit of the invariant axis absorbs the oscillation entirely, leaving no residual to read; and the longitudinal spin component is insensitive to the relevant knobs off-magic. The magic momentum is the unique operating point at which the dominant first-order effect is converted into a harmless constant offset — which is also why the estimator of Sec. II B functions there.

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